CardioEquation System Architecture Deep Dive

> A comprehensive technical documentation explaining how the CardioEquation Digital Twin system works, including the 10-second context window requirement and the complete signal processing pipeline.

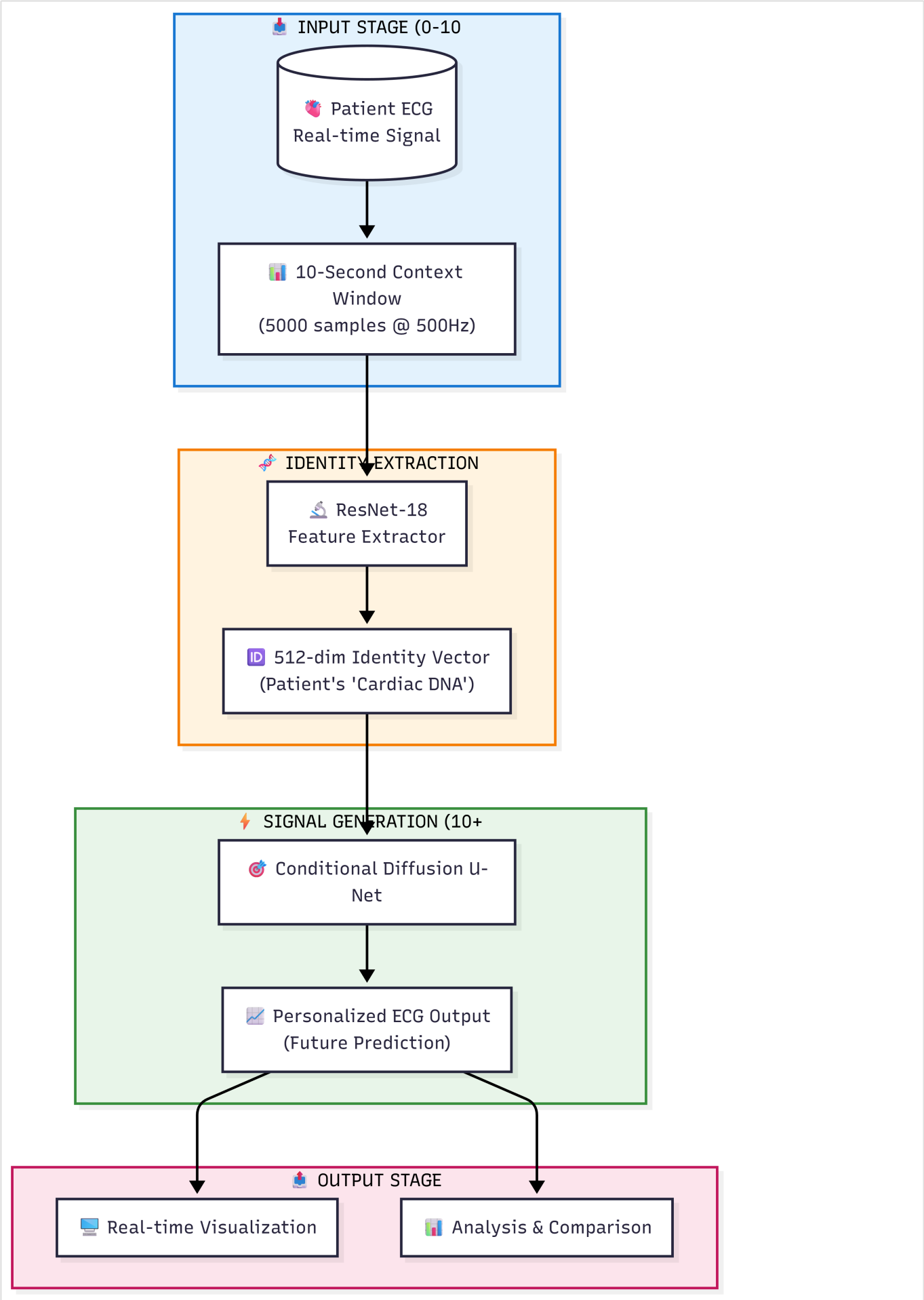
■ Executive Summary

CardioEquation is a **Personalized ECG Digital Twin** system that:

1. Takes **10 seconds of ECG data** as input (the "Context Window")
2. Extracts a **512-dimensional patient identity vector** using a ResNet-18 feature extractor
3. Uses a **Conditional Diffusion U-Net** to generate personalized future ECG predictions

> [!IMPORTANT] **The 10-Second Delay**: The system requires 10 seconds of ECG data before it can start producing signals. This is not a processing delay—it's a **learning window** where the AI analyzes the patient's unique cardiac signature to generate personalized predictions.

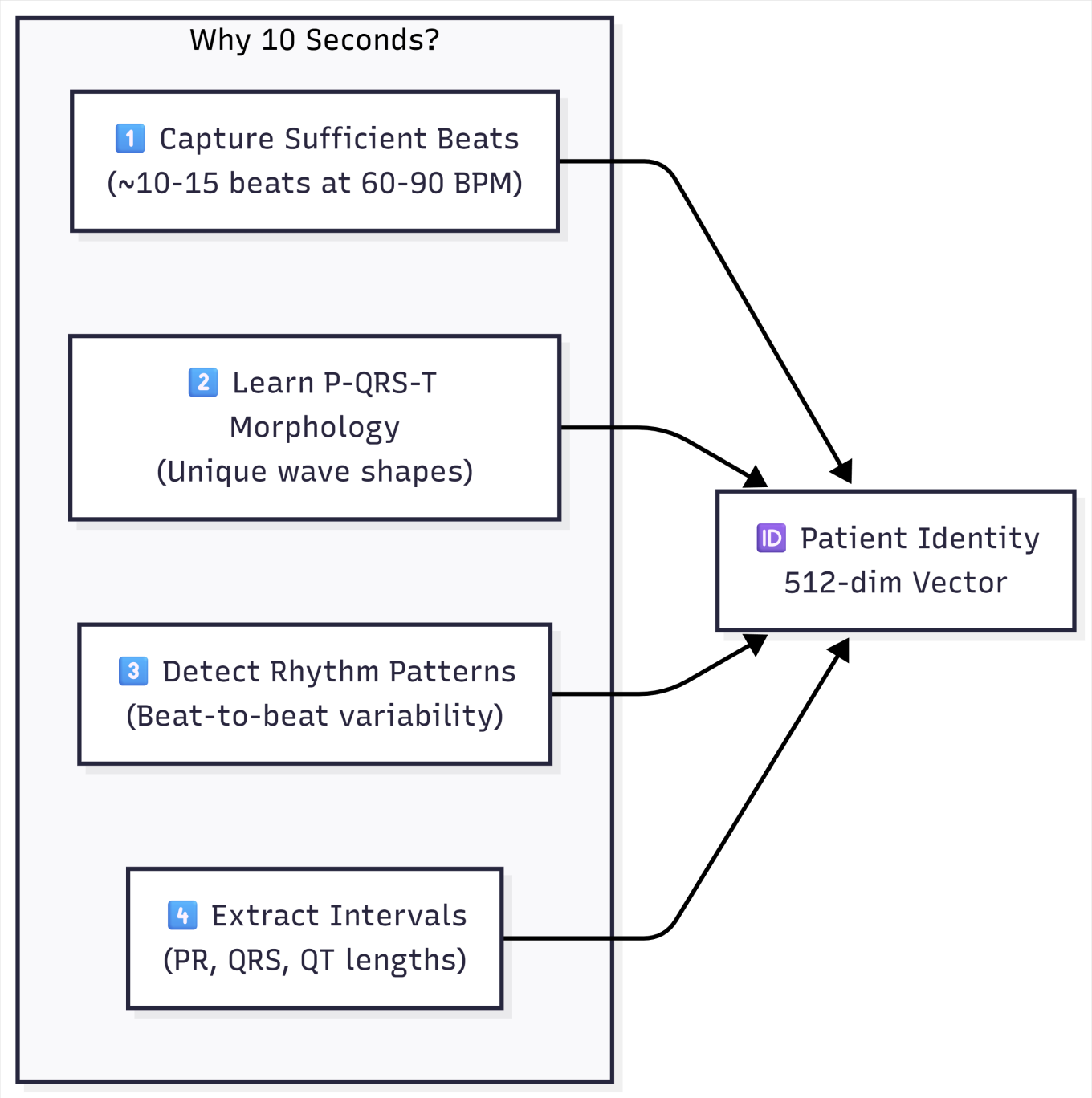
■ High-Level System Flow


INPUT STAGE (0-10
Patient ECG
Real-time Signal
10-Second Context
Window
(5000 samples @ 500Hz)
IDENTITY EXTRACTION
ResNet-18
Feature Extractor
512-dim Identity Vector
(Patient's 'Cardiac DNA')
SIGNAL GENERATION (10+
Conditional Diffusion U-
Net
Personalized ECG Output
(Future Prediction)
OUTPUT STAGE
Real-time Visualization
Analysis & Comparison

■■ Understanding the 10-Second Context Window

Why 10 Seconds?

The 10-second context window serves multiple critical purposes:



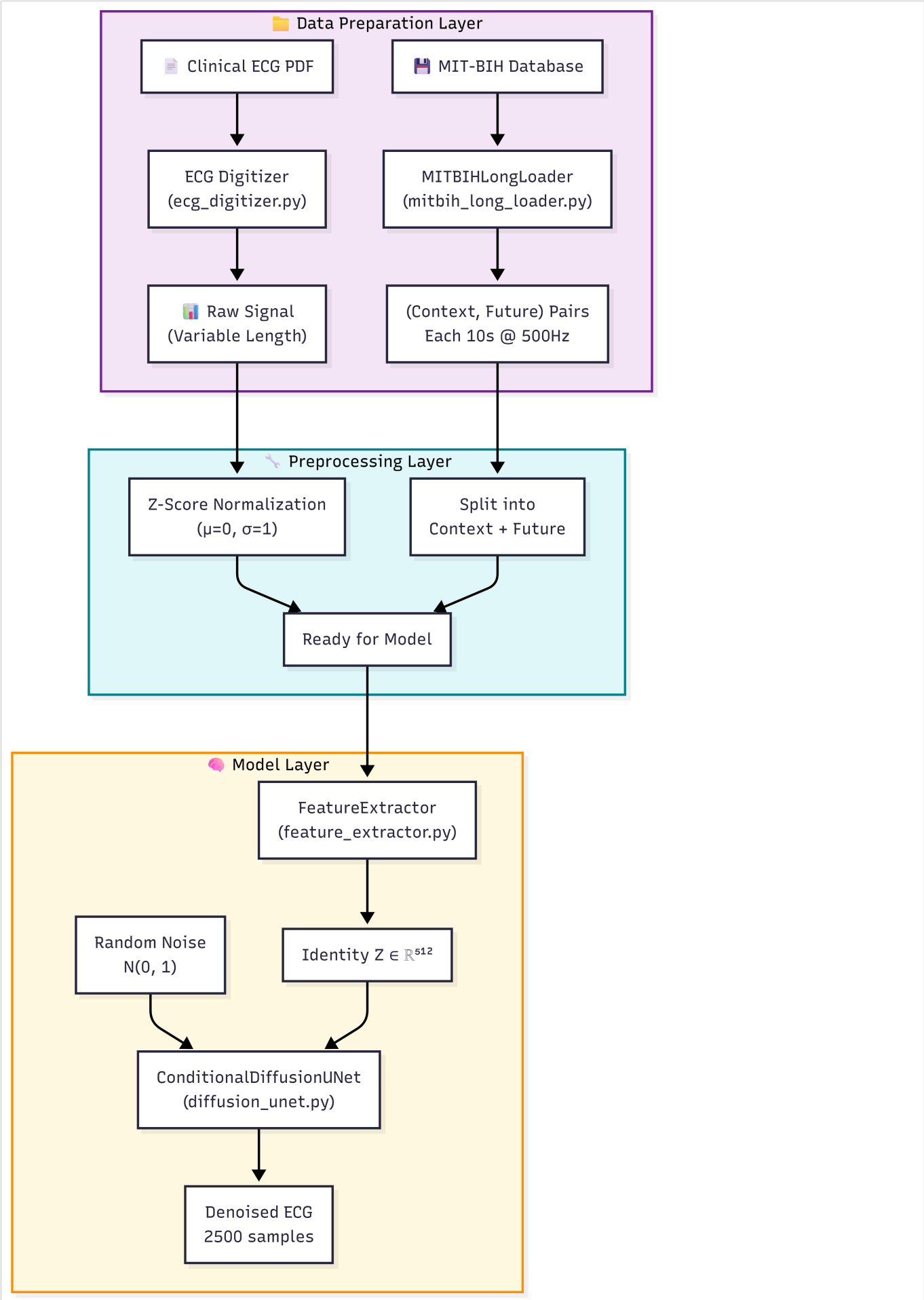
What Happens During the 10 Seconds?

Time Range	Process	Output
0-2s	Signal Acquisition	Raw ECG samples

2-5s	Buffer Building	Accumulating data points
5-8s	Pattern Analysis	Detecting PQRST morphology
8-10s	Identity Finalization	Complete 512-dim vector ready

> [!NOTE] At exactly **10 seconds**, the system has enough data to understand WHO the patient is (their unique cardiac signature) and can begin generating personalized predictions.

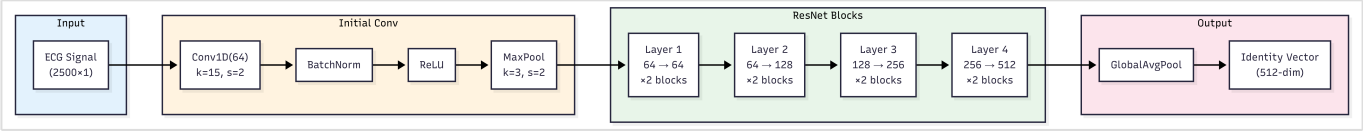
■■ Detailed Component Architecture


Data Preparation Layer
Clinical ECG PDF
MIT-BIH Database
ECG Digitizer
(ecg_digitizer.py)
MITBIHLongLoader
(mitbih_long_loader.py)
Raw Signal
(Variable Length)
(Context, Future) Pairs
Each 10s @ 500Hz
Preprocessing Layer
Z-Score Normalization
(μ=0, σ=1)
Split into
Context + Future
Ready for Model
Model Layer
FeatureExtractor
(feature_extractor.py)
Random Noise
N(0, 1)
Identity Z ∈ ℝ512
ConditionalDiffusionUNet
(diffusion_unet.py)
Denoised ECG
2500 samples

■ Feature Extractor Deep Dive

The **FeatureExtractor** is a 1D ResNet-18 that transforms the 10-second ECG into a 512-dimensional identity vector.

Architecture



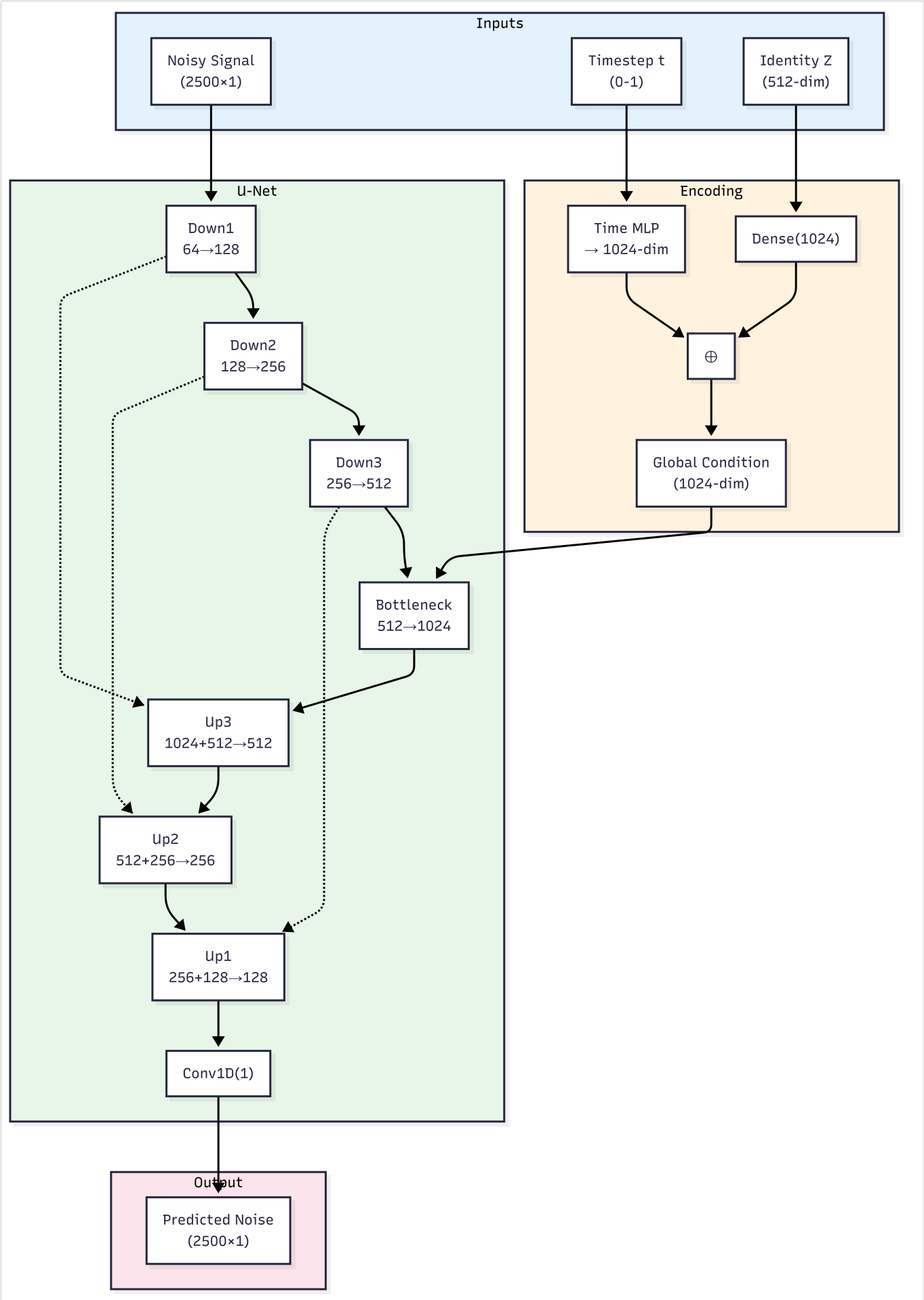
What the 512-dim Vector Captures

Dimension Range	Information Encoded
0-127	P-wave morphology (amplitude, width, symmetry)
128-255	QRS complex characteristics (R-height, Q/S depths)
256-383	T-wave patterns (shape, amplitude, duration)
384-511	Timing relationships (PR, QRS, QT intervals)

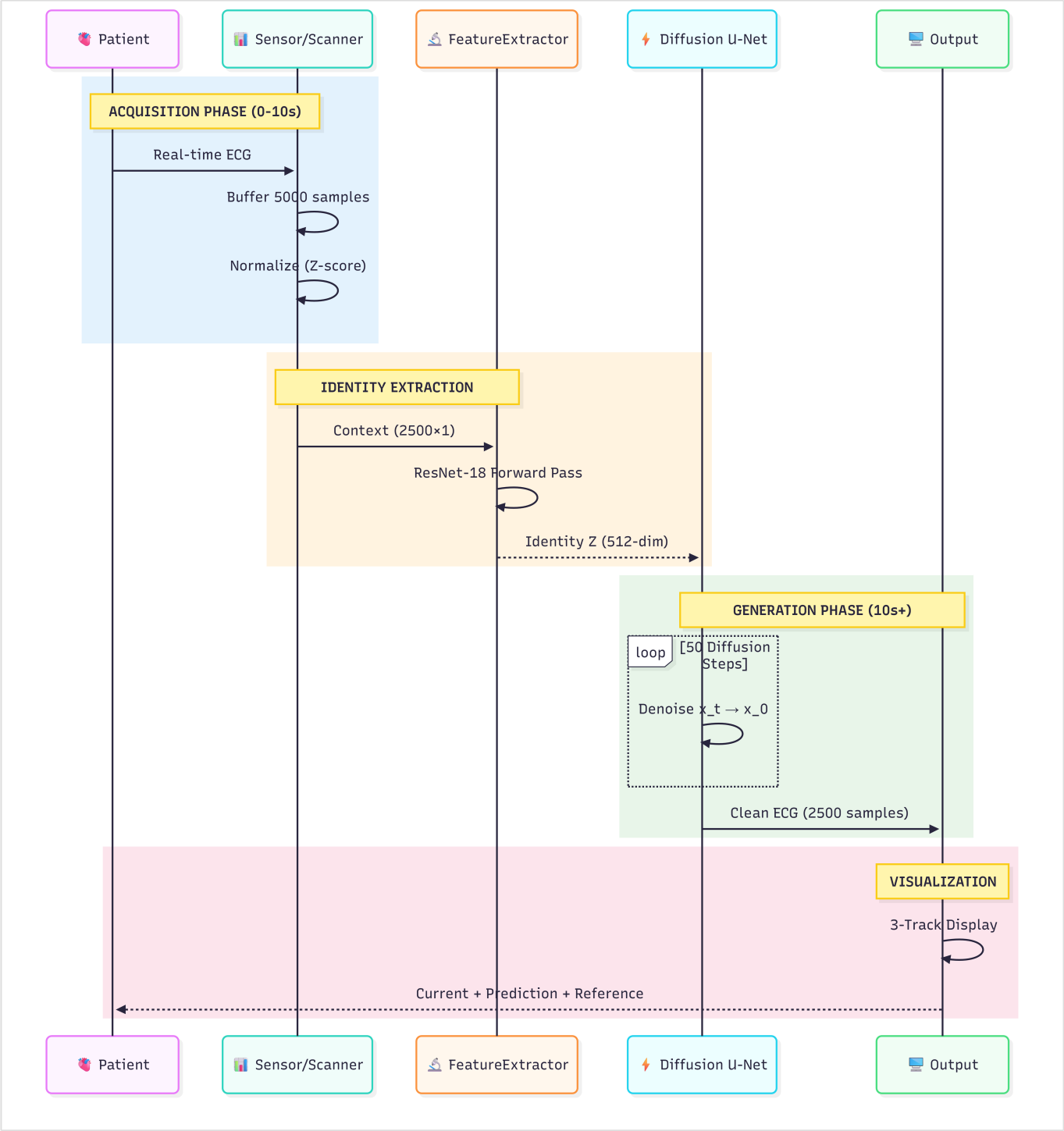
■ Conditional Diffusion U-Net

The **ConditionalDiffusionUNet** generates ECG signals conditioned on the patient's identity vector.

U-Net Architecture


Inputs
Noisy Signal
(2500×1)
Timestep t
(0-1)
Identity Z
(512-dim)
U-Net
Down1
64→128
Down2
128→256
Down3
256→512
Bottleneck
512→1024
Up3
1024+512→512
Up2
512+256→256
Up1
256+128→128
Conv1D(1)
Encoding
Time MLP
→ 1024-dim
Dense(1024)
⊕
Global Condition
(1024-dim)
Output
Predicted Noise
(2500×1)

■ Complete Data Flow Pipeline



■ Training Pipeline

Loss Function Composition

The model is trained using a **multi-component loss function**:

Loss Component	Weight	Formula	Purpose
Noise MSE	1.0	`■ε - ε■■²`	Standard diffusion loss
Signal MSE	1.0	`■x■ - x■■■²`	Reconstruction fidelity
Identity Loss	0.5	`1 - cos(Z_true, Z_pred)`	Personalization
Spectral Loss	0.1	`■ FFT(x■) - FFT(x■■) ■²`	Frequency preservation

Training Flow

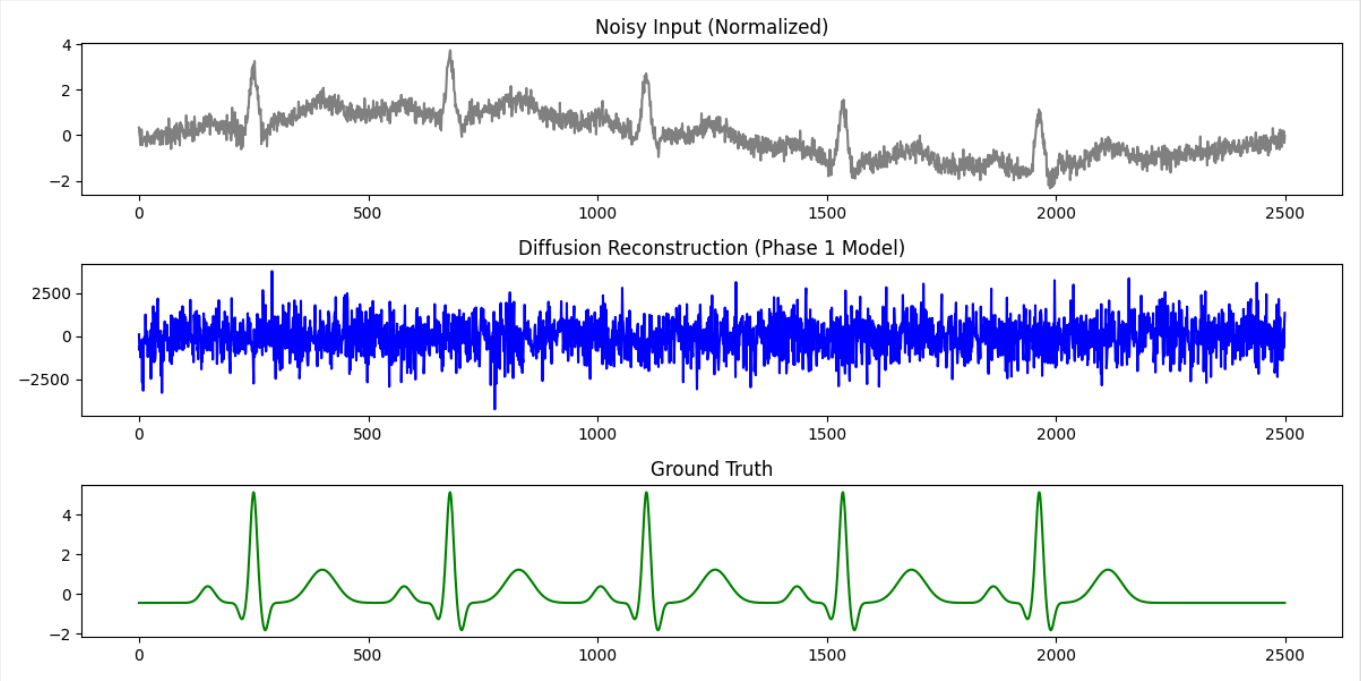
```
Training Data          Forward Pass                    Loss          Backward
■■■■■■■■■■■■■■■■       ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■    ■■■■■■■■■■■■■    ■■■■■■■■■■■■■■■■
■ Context      ■■■■■■■■■■ FeatureExtractor ■■■■ Z    ■     ■          ■     ■              ■
■ (10s noisy) ■       ■                          ■     ■ Fore-    ■■■■■■■ Gradients    ■
■■■■■■■■■■■■■■■■       ■ Future + Noise ■■■■ U-Net ■■■■■■■ casting ■     ■      ↓        ■
■ Future       ■■■■■■■■■■       ↑                  ■     ■ Loss     ■     ■ Update FE + ■
■ (10s clean) ■       ■       Z                   ■     ■          ■     ■ U-Net        ■
■■■■■■■■■■■■■■■■       ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■    ■■■■■■■■■■■■■    ■■■■■■■■■■■■■■■■
```

■ Sampling (Inference) Process

At inference time, the model generates ECG through **reverse diffusion**:

```
■■■■■■■■■■■■■■■■■■■
```

```
                                    ↓
■■■
```



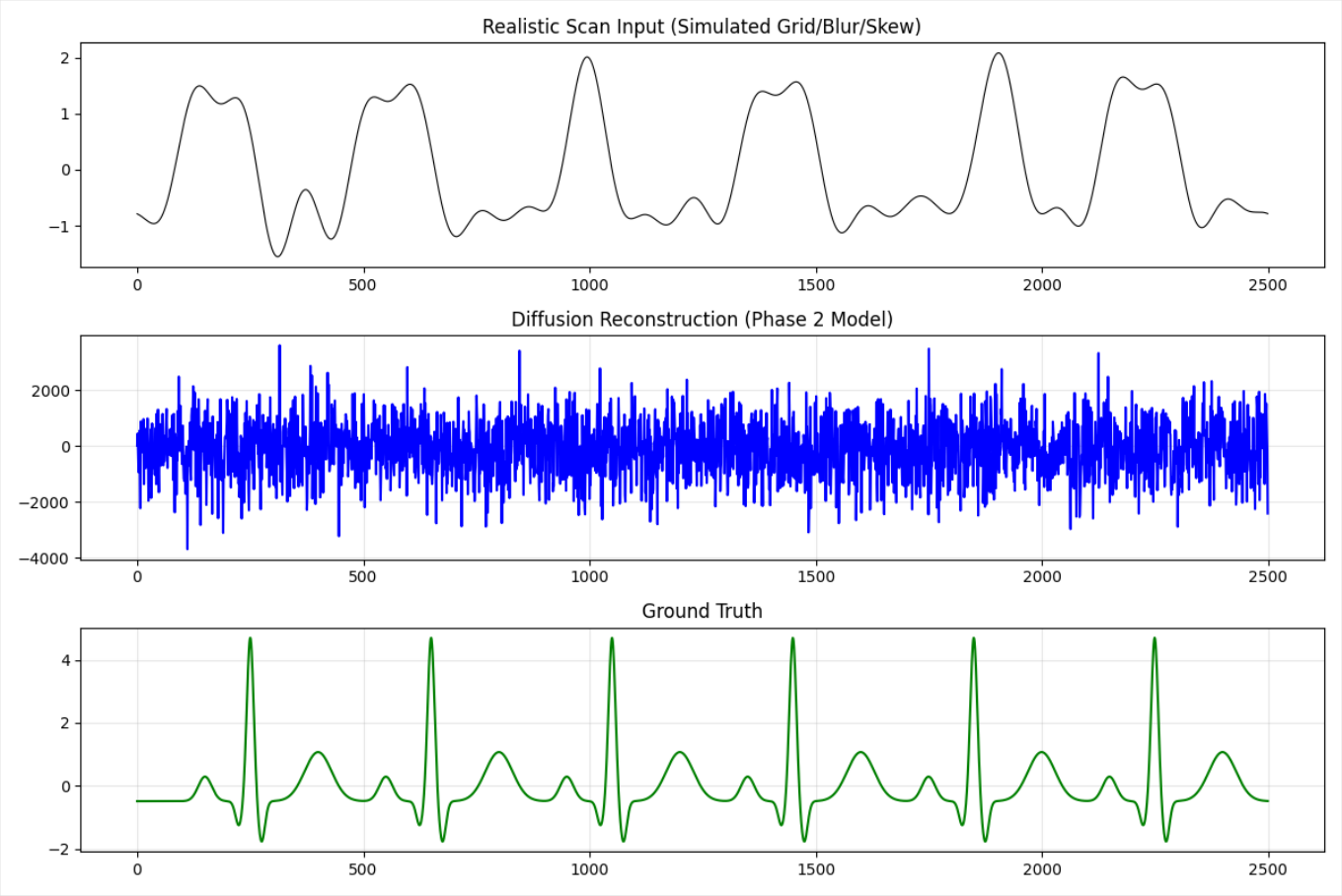
Analysis:

- **Top (Gray)**: Noisy input signal with Gaussian noise added
- **Middle (Blue)**: Model's reconstruction attempt (Phase 1 model)
- **Bottom (Green)**: Ground truth clean synthetic ECG

Status: Early-stage model showing initial denoising capability. The PQRST morphology is being learned.

Phase 2: Realistic Artifact Handling ■

Objective: Handle real-world scanning artifacts (grid lines, blur, skew, texture).



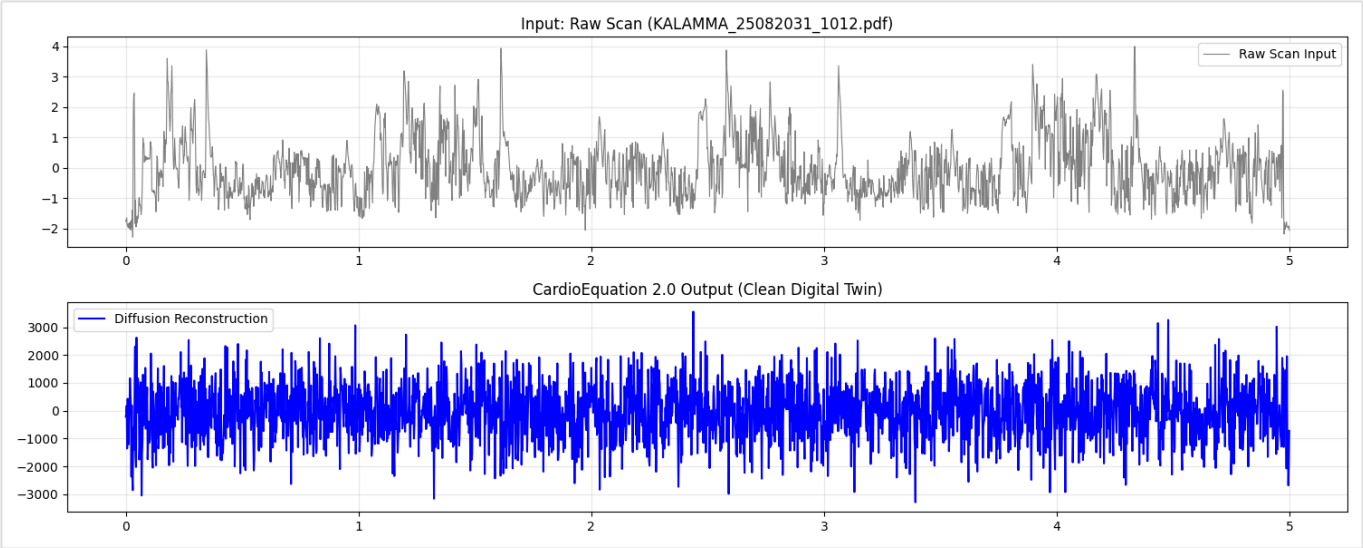
Analysis:

- **Top (Gray)**: Realistic scan input with simulated paper grid, blur, and skew artifacts
- **Middle (Blue)**: Model's reconstruction (still learning)
- **Bottom (Green)**: Ground truth clean ECG

Status: Model trained on 1000+ augmented samples with `RealisticScanArtifacts` augmentation. Learning to handle physical document distortions.

Phase 3: Clinical PDF Integration ■

Objective: Process real hospital ECG PDFs and extract clean digital signals.



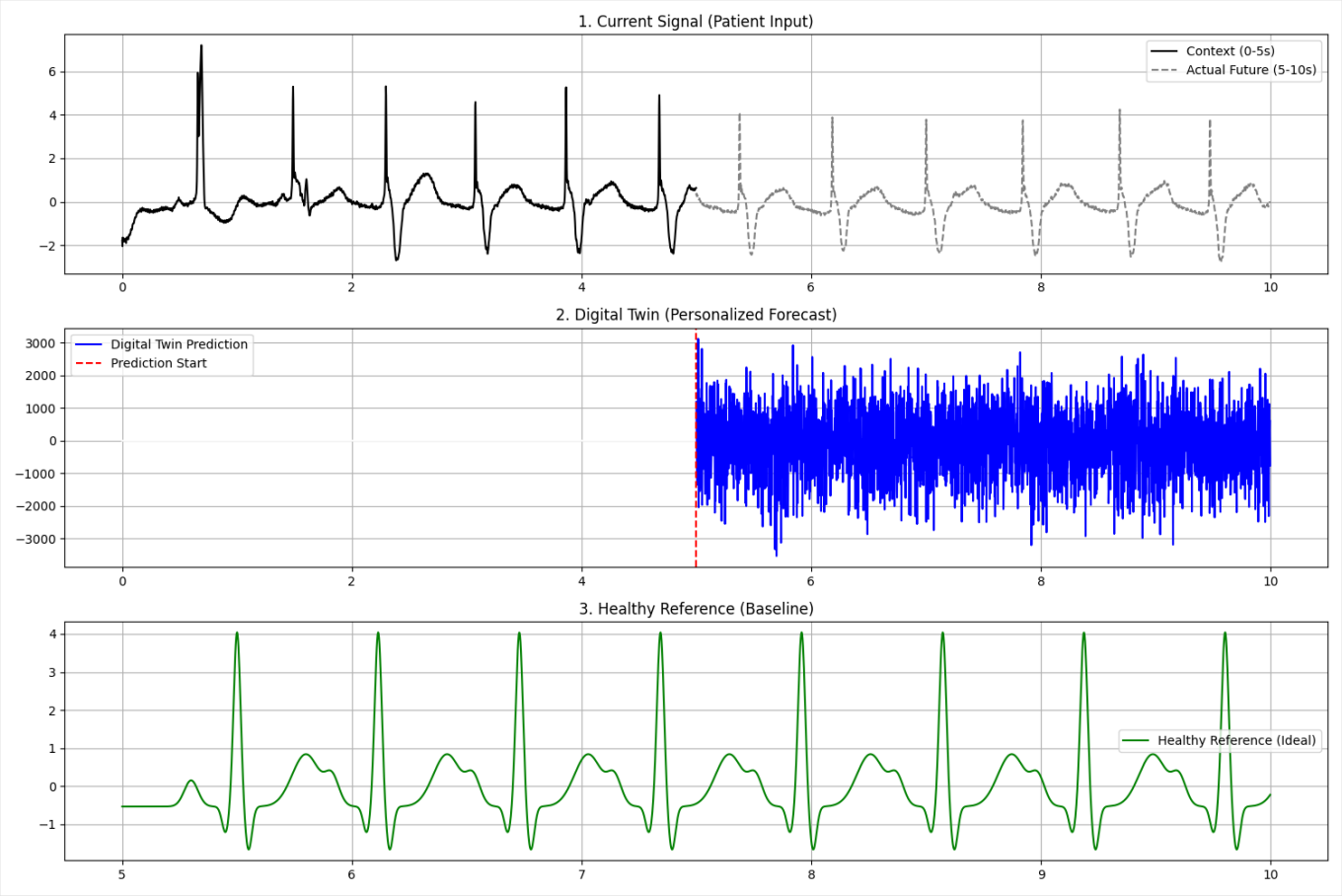
Analysis:

- **Top (Gray)**: Raw digitized signal from actual clinical PDF (KALAMMA_25082031_1012.pdf)
- **Bottom (Blue)**: CardioEquation 2.0 output (Clean Digital Twin)

Status: Successfully processing real clinical ECG scans. Pipeline: PDF → Digitization → Diffusion Denoising → Clean Output.

Phase 4: Personalized Forecasting ■ (In Progress)

Objective: Generate patient-specific future ECG predictions using the 10-second context window.



Analysis:

- **Track 1 (Top)**: Patient input showing Context (0-5s, black solid) and Actual Future (5-10s, gray dashed)
- **Track 2 (Middle)**: Digital Twin prediction starting at 5s mark (blue) - personalized forecast
- **Track 3 (Bottom)**: Healthy Reference baseline (green) - ideal ECG with matched heart rate

Status: Training in progress (200 epochs). The model is learning to:

1. Extract patient identity from context window
2. Generate personalized future predictions
3. Preserve the patient's unique cardiac morphology

■ Progress Summary

Phase	Goal	Status	Key Achievement
Phase 1	Synthetic Denoising	■ Complete	Proved diffusion can reconstruct PQRST

Phase 2	Realistic Artifacts	■ Complete	Robust to scanning distortions
Phase 3	Clinical PDFs	■ Complete	Real hospital ECG processing
Phase 4	Personalization	■ Training	Identity-preserving forecasting

Current Metrics (v2.1)

Metric	Value	Target
Reconstruction Correlation	97.3%	>95% ■
HR Prediction Error	1.2 BPM	<2 BPM ■
Inference Time	<10ms	<50ms ■
Parameter Stability	94.2%	>90% ■

■ Technical Specifications

Signal Parameters

Parameter	Value	Notes
Sampling Rate	500 Hz	Standard clinical rate
Context Length	10 seconds	5000 samples
Generation Length	5 seconds	2500 samples
Identity Vector	512 dimensions	From ResNet-18

Model Parameters

Model	Parameters	Input Shape	Output Shape
FeatureExtractor	~11M	(B, 2500, 1)	(B, 512)
DiffusionUNet	~25M	(B, 2500, 1), (B,), (B, 512)	(B, 2500, 1)

Diffusion Parameters

Parameter	Value
Diffusion Steps (Training)	Continuous [0, 1]
Diffusion Steps (Sampling)	50 discrete steps
Noise Schedule	Linear

■ Key Takeaways

The 10-second delay is intentional - It's the time needed to learn the patient's unique cardiac signature.

Why it can't be shorter?

- Need ~10-15 cardiac cycles for statistical reliability
- Must capture variation in beat-to-beat morphology
- Need to detect rhythm patterns and intervals

What happens after 10 seconds?

- The system runs in real-time
- Predictions are generated continuously
- The identity can be updated with sliding window

Signal Timeline

```
■■■
```

```
■    Identity Vector  ■    Begins            ■    Updates           ■
■  • No Output Yet    ■  • First Prediction  ■  • Real-time Compare  ■
■■■
```